



GIT

Introdução

PAULO R T CÂNDIDO

Git – Comandos Principais

[prt candidato/cicd: Material aula CICD](#)

`mkdir testeGit && cd testeGit`

Transformar pasta em um repositório git

`git init`

Mostrar o nome do branch em que se está no momento

`git branch --show-current`

Renomeia branch para main

`git branch -m main`

Listar as configurações globais

`git config --global --list`

Consultar configurações globais

`git config --global user.email`

Alterar configurações globais

`git config --global user.name "Paulo Candido"`

Git – Comandos Principais

Configurar repositório remoto no GITHUB (necessário criar manualmente no Github)

```
git remote add origin https://github.com/prtcandido/testeGit.git
```

Verificar configuração de armazenamento remoto

```
git remote -v
```

Alterar URL de armazenamento remoto

```
git remote set-url origin https://github.com/prtcandido/gitaula1.git
```

Remover armazenamento remoto

```
git remote remove origin
```

Reconfigurar <<<<

```
git remote add origin https://github.com/prtcandido/testeGit.git
```

Git – Comandos Principais

Criar estes arquivos

```
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatec
Praia
Grande

prtcandido@PauloAcer:~/testeGit$ cat a.log
teste
teste
x
```

Listar os arquivos cujo versionamento está sendo controlado

```
prtcandido@PauloAcer:~/testeGit$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    a.log
    a.txt

nothing added to commit but untracked files present (use "git add" to track)
```

Git – Comandos Principais

Criar arquivo .gitignore, todos os arquivos log serão ignorados
nano .gitignore

```
GNU nano 7.2
*.log
```

Todos os arquivos log são ignorados

```
prtcandido@PauloAcer:~/testeGit$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  .gitignore
  a.txt

nothing added to commit but untracked files present (use "git add" to track)
```

Git – Comandos Principais

Adicionar arquivo no stage para commit

```
prtcandido@PauloAcer:~/testeGit$ git add a.txt
prtcandido@PauloAcer:~/testeGit$ git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   a.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
```

Retirar arquivo do stage

```
prtcandido@PauloAcer:~/testeGit$ git rm --cached a.txt
rm 'a.txt'
prtcandido@PauloAcer:~/testeGit$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        a.txt

nothing added to commit but untracked files present (use "git add" to track)
```

Git – Comandos Principais

Adiciona todos os arquivos novos e modificados no stage para commit (pacote de alterações)

```
prtcandido@PauloAcer:~/testeGit$ git add .
prtcandido@PauloAcer:~/testeGit$ git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   .gitignore
    new file:   a.txt

prtcandido@PauloAcer:~/testeGit$ git commit -m primeiro
[main (root-commit) 6f4b108] primeiro
 2 files changed, 5 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 a.txt
```

Listar commits realizados

```
prtcandido@PauloAcer:~/testeGit$ git log --oneline
6f4b108 (HEAD -> main) primeiro
```

Git – Comandos Principais

Primeiro push para remoto origin do branch main (necessário criar repositório no github)

```
prtcandido@PauloAcer:~/testeGit$ git push -u origin main
Username for 'https://github.com': prtcandido
Password for 'https://prtcandido@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (4/4), 283 bytes | 283.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/prtcandido/testeGit.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
prtcandido@PauloAcer:~/testeGit$ |
```

Diferença de conteúdo entra a versão atual de a.txt e a versão do último commit.

```
prtcandido@PauloAcer:~/testeGit$ nano a.txt
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatec
Praiaaa
Grande

prtcandido@PauloAcer:~/testeGit$ git diff a.txt
diff --git a/a.txt b/a.txt
index 7ce909d..8132ede 100644
--- a/a.txt
+++ b/a.txt
@@ -1,4 +1,4 @@
 Fatec
-Praia
+Praiaaaa
 Grande
```

Diferença com o arquivo do commit 6f4b108

```
prtcandido@PauloAcer:~/testeGit$ git diff 6f4b108 a.txt
diff --git a/a.txt b/a.txt
index 7ce909d..8132ede 100644
--- a/a.txt
+++ b/a.txt
@@ -1,4 +1,4 @@
 Fatec
-Praia
+Praiaaaa
 Grande
```


Git – Comandos Principais

Recupera versão anterior de um commit (pacote de alterações) específico.

```
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatec
Praiaaa
Grande

prtcandido@PauloAcer:~/testeGit$ git restore --source 6f4b108 a.txt
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatec
Praia
Grande

prtcandido@PauloAcer:~/testeGit$ |
```

Comparar com arquivo que está em stage

git diff --cached a.txt

Git – Comandos Principais

Criar dois branches (cópias!!!): b1 baseado na main; b2 baseado no commit 6f4b108

```
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatec
Praia
Grande

prtcandido@PauloAcer:~/testeGit$ git branch b1
prtcandido@PauloAcer:~/testeGit$ git checkout -b b2 6f4b108
Switched to a new branch 'b2'
```

Modificar o arquivo a.txt de forma diferente nos dois branches e fazer commit das mudanças.

```
prtcandido@PauloAcer:~/testeGit$ git checkout b1
Switched to branch 'b1'
prtcandido@PauloAcer:~/testeGit$ nano a.txt
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatecccc
Praia
Grande

prtcandido@PauloAcer:~/testeGit$ git add .
prtcandido@PauloAcer:~/testeGit$ git commit -m b1_1
[b1 6a79844] b1_1
1 file changed, 1 insertion(+), 1 deletion(-)
prtcandido@PauloAcer:~/testeGit$ git checkout b2
Switched to branch 'b2'
prtcandido@PauloAcer:~/testeGit$ nano a.txt
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatec
Praia
Grandeeee

prtcandido@PauloAcer:~/testeGit$ git add .
prtcandido@PauloAcer:~/testeGit$ git commit -m b2_1
[b2 c95e662] b2_1
1 file changed, 1 insertion(+), 1 deletion(-)
prtcandido@PauloAcer:~/testeGit$ |
```

Git – Comandos Principais

Mesclar a alterações feitas nos branches b1 e b2 no main.

```
prtcandido@PauloAcer:~/testeGit$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
prtcandido@PauloAcer:~/testeGit$ git merge b1
Updating 6f4b108..6a79844
Fast-forward
 a.txt | 2 +-
 1 file changed, 1 insertion(+), 1 deletion(-)
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatecccc
Praia
Grande

prtcandido@PauloAcer:~/testeGit$ git merge b2
Auto-merging a.txt
Merge made by the 'ort' strategy.
 a.txt | 2 +-
 1 file changed, 1 insertion(+), 1 deletion(-)
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatecccc
Praia
Grande
```

Git – Comandos Principais

Gerar conflito para merge alterando a mesma linha de a.txt nos branches main e b1

```
prtcandido@PauloAcer:~/testeGit$ git branch --show-current
main
prtcandido@PauloAcer:~/testeGit$ nano a.txt
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatecccc
Praiaxxxxxxxxx
Grandegee

prtcandido@PauloAcer:~/testeGit$ git checkout b1
error: Your local changes to the following files would be overwritten by checkout:
a.txt
Please commit your changes or stash them before you switch branches.
Aborting
prtcandido@PauloAcer:~/testeGit$ git add .
prtcandido@PauloAcer:~/testeGit$ git commit -m main_1
[main e7e666e] main_1
1 file changed, 1 insertion(+), 1 deletion(-)
prtcandido@PauloAcer:~/testeGit$ git checkout b1
Switched to branch 'b1'
prtcandido@PauloAcer:~/testeGit$ nano a.txt
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatecccc
Praiaaaaaaaaaa
Grande

prtcandido@PauloAcer:~/testeGit$ git add .
prtcandido@PauloAcer:~/testeGit$ git commit -m b1_2
[b1 df185ca] b1_2
1 file changed, 1 insertion(+), 1 deletion(-)
prtcandido@PauloAcer:~/testeGit$ git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 4 commits.
(use "git push" to publish your local commits)
prtcandido@PauloAcer:~/testeGit$ git merge b1
Auto-merging a.txt
CONFLICT (content): Merge conflict in a.txt
Automatic merge failed; fix conflicts and then commit the result.
prtcandido@PauloAcer:~/testeGit$ |
```

```
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatecccc
<<<<<< HEAD
Praiaxxxxxxxxx
Grandegee
=====
Praiaaaaaaaaaa
Grande
>>>>>> b1
```

ours VS theirs

Se houver um conflito de código, você diz ao Git quem deve "vencer" automaticamente.

- git merge -Xours feature: Em caso de conflito, mantém a versão da sua branch atual (**main**).
- git merge -Xtheirs feature: Em caso de conflito, aceita as mudanças da branch que está vindo (**feature**).

Git – Comandos Principais

```
prtcandido@PauloAcer:~/testeGit$ git branch -a
b1
b2
* main
remotes/origin/main
prtcandido@PauloAcer:~/testeGit$ git checkout -b b3
Switched to a new branch 'b3'
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatecccc
Praiaxxxxxxxxx
Grandegee
prtcandido@PauloAcer:~/testeGit$ nano a.txt
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatecccc#####
Praiaxxxxxxxxx$$$$$$$$$$$$$$$
Grandegee@@@@@@@@@@@@@@@@
prtcandido@PauloAcer:~/testeGit$ git add .
prtcandido@PauloAcer:~/testeGit$ git commit -m b3_1
[b3 5ccf9a0] b3_1
1 file changed, 3 insertions(+), 3 deletions(-)
prtcandido@PauloAcer:~/testeGit$ git push -u origin b3
Username for 'https://github.com': prtcandido
Password for 'https://prtcandido@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 306 bytes | 306.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'b3' on GitHub by visiting:
remote: https://github.com/prtcandido/testeGit/pull/new/b3
remote:
To https://github.com/prtcandido/testeGit.git
* [new branch]      b3 -> b3
branch 'b3' set up to track 'origin/b3'.
prtcandido@PauloAcer:~/testeGit$
```

Feito push do novo Branch (b3) é possível gerar um Pull Request no Github

Git – Pull Request

Open a pull request

Create a new pull request by comparing changes across two branches. If you need to, you can also [compare across forks](#). [Learn more about diff comparisons here](#).

 base: main  compare: b3 ✓ Able to merge. These branches can be automatically merged.



Add a title *

Aula sobre GIT 

Add a description

Write Preview

H B I        @   

Apenas um teste

 Markdown is supported  Paste, drop, or click to add files

Reviewers 

No reviews

Assignees 

No one—[assign yourself](#)

Labels 

None yet

Projects 

None yet

Milestone 

No milestone

Development

Use [Closing keywords](#) in the description to automatically close issues

Helpful resources

Create pull request 

Git – Pull Request

Aula sobre GIT #1 ✎

✓ Ready to merge

Code ▾

🔗 Open prtcardido wants to merge 1 commit into main from b3 📄

💬 Conversation 0

🔑 Commits 1

📄 Checks 0

📄 Files changed 1

+3 -3 🟢🔴🔴🔴



prtcardido commented 1 minute ago

Owner ⋮

Apenas um teste



5ccf9a0

Pronto para fazer o Merge



No conflicts with base branch

Merging can be performed automatically.

Merge pull request ▾

You can also merge this with the command line. [View command line instructions.](#)

Still in progress? [Convert to draft](#)



Add a comment

Write

Preview

H B I ☰ <> 🔗 ☰ ☰ ☰ 📎 @ 📄 ↩ ☑

Add your comment here...

Reviewers



No reviews

Still in progress? [Convert to draft](#)

Assignees



No one—[assign yourself](#)

Labels



None yet

Projects



None yet

Milestone



No milestone

Development



Successfully merging this pull request may close these issues.

None yet

Notifications

[Customize](#)

Git – Pull Request

https://github.com/prtcandido/testeGit/settings/branches

prtcandido / testeGit

Code Issues Pull requests 1 Actions Projects Wiki Security Insights **Settings**

General

Access

Collaborators

Moderation options

Code and automation

Branches

Tags

Rules

Actions

Models [Preview](#)

Webhooks

Branch protection rules

Classic branch protections have not been configured
Define branch rules to disable force pushing, prevent branches from being deleted, or require pull requests b
Learn more about [repository rules](#) and [protected branches](#).

[Add branch ruleset](#) **[Add classic branch protection rule](#)**

Git – Pull Request

General

Access

- Collaborators
- Moderation options

Code and automation

- Branches**
- Tags
- Rules
- Actions
- Models
- Webhooks
- Copilot
- Environments
- Codespaces
- Pages

Security

- Advanced Security
- Deploy keys
- Secrets and variables

Integrations

Branch protection rule

Protect your most important branches

[Branch protection rules](#) define whether collaborators can delete or force push to the branch and set requirements for any pushes to the branch, such as passing status checks or a linear commit history.

[Your GitHub Free plan](#) can only enforce rules on its public repositories, like this one.

Branch name pattern *

main

Protect matching branches

- ☒ **Require a pull request before merging**
When enabled, all commits must be made to a non-protected branch and submitted via a pull request before they can be merged into a branch that matches this rule.
- ☒ **Require approvals**
When enabled, pull requests targeting a matching branch require a number of approvals and no changes requested before they can be merged.
Required number of approvals before merging: 1
- ☐ **Dismiss stale pull request approvals when new commits are pushed**
New reviewable commits pushed to a matching branch will dismiss pull request review approvals.
- ☐ **Require review from Code Owners**
Require an approved review in pull requests including files with a designated code owner.
- ☐ **Require approval of the most recent reviewable push**

Git – Pull Request

The screenshot shows a GitHub Pull Request (PR) page for the repository 'prtcandido / testeGit'. The PR is titled 'Aula sobre GIT #1' and is in the 'Awaiting approval' state. It shows a merge from branch 'b3' to 'main' with 1 commit. The PR has 1 conversation, 1 commit, 0 checks, and 1 file changed. A comment by 'prtcandido' from last week says 'Apenas um teste'. The PR is self-assigned by 'prtcandido' 20 minutes ago. The right sidebar shows the 'Reviewers' section with a message: 'No reviews—at least 1 approving review is required. Still in progress? [Convert to draft](#)'. The bottom of the page shows two error messages: 'Review required' and 'Merging is blocked', both stating 'At least 1 approving review is required by reviewers with write access.'.

https://github.com/prtcandido/testeGit/pull/1

prtcandido / testeGit

de Issues Pull requests 1 Actions Projects Wiki Security Insights Settings

Aula sobre GIT #1

Awaiting approval Code

Open prtcandido wants to merge 1 commit into main from b3

Conversation 0 Commits 1 Checks 0 Files changed 1 +3 -3

prtcandido commented last week

Apenas um teste

b3_1 5ccf9a0

prtcandido self-assigned this 20 minutes ago

Review required
At least 1 approving review is required by reviewers with write access.

Merging is blocked
At least 1 approving review is required by reviewers with write access.

Reviewers
No reviews—at least 1 approving review is required.
Still in progress? [Convert to draft](#)

Assignees
prtcandido

Labels
None yet

Projects
None yet

Milestone
No milestone

Development

Git – Pull Request

Aula sobre GIT #1

● Awaiting review

 Open [prtcandido](#) wants to merge 1 commit into `main` from `b3` 

 Conversation 0  Commits 1  Checks 0  Files changed 1



[prtcandido](#) commented [last week](#)

Owner

...

Apenas um teste



 [prtcandido](#) self-assigned this [31 minutes ago](#)

 [prtcandido](#) added the `bug` label [4 minutes ago](#)

Em **CI/CD e Automação**, as labels são gatilhos. Por exemplo, se um Pull Request recebe a label `deploy-to-prod`, um workflow pode disparar o deploy automaticamente.

[5ccf9a0](#)

Reviewers

No reviews—at least 1 approved review is required to merge
Still in progress? [Convert to draft](#)

Assignees

 [prtcandido](#)

Labels

`bug`

Projects

None yet